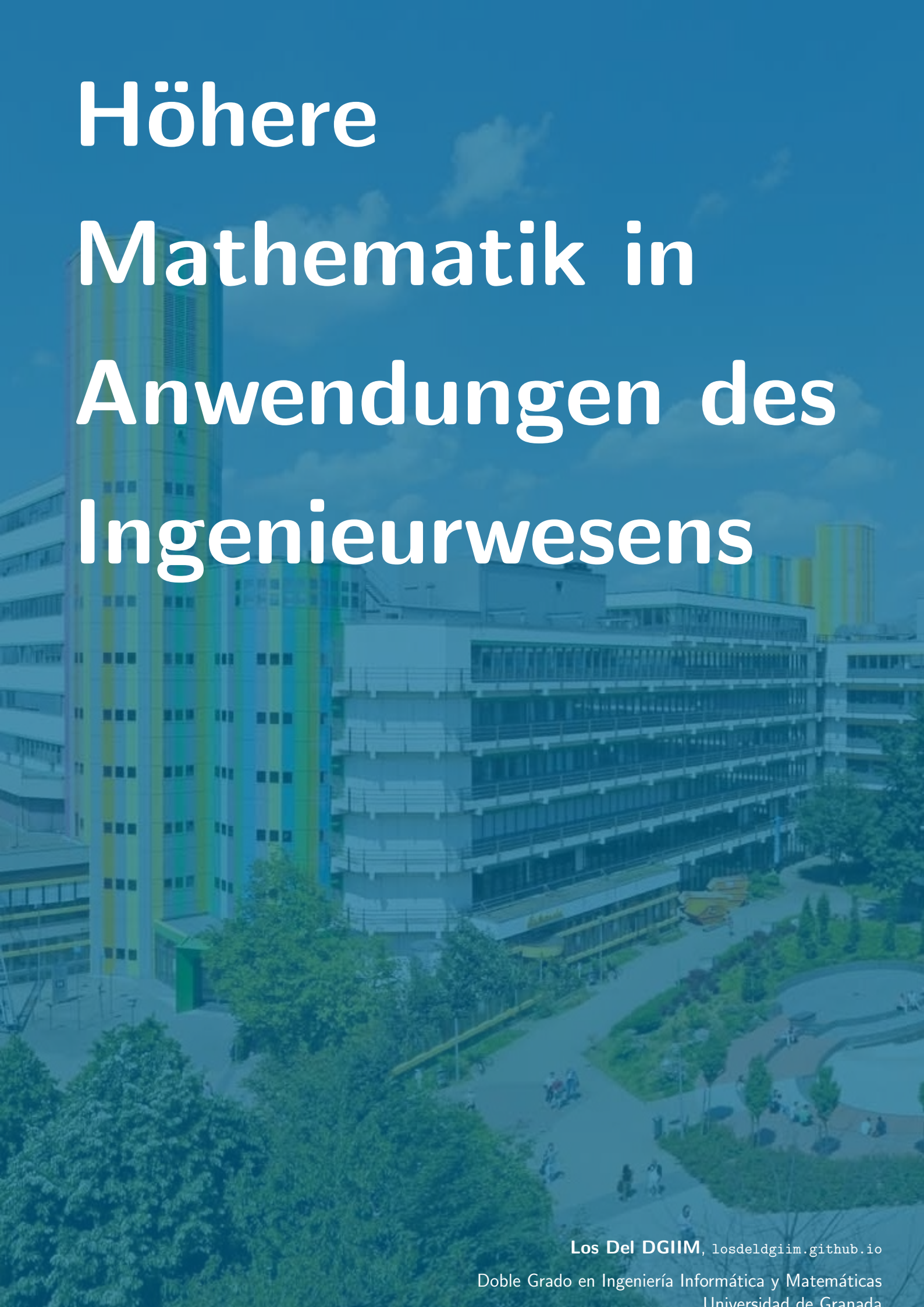


Höhere Mathematik in Anwendungen des Ingenieurwesens



Los Del DGIIM, losdeldgiim.github.io

Doble Grado en Ingeniería Informática y Matemáticas
Universidad de Granada

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Höhere Mathematik in Anwendungen des Ingenieurwesens

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1. Introduction to MATLAB

MATLAB is a high-level programming language and environment used for numerical computing, data analysis, and visualization. It is widely used in engineering, science, and mathematics for tasks such as matrix manipulation, algorithm development, and data visualization. A license is required to use MATLAB, and it is commonly used in academic and research settings. However, there are alternatives to MATLAB, such as Octave and Scilab, which are open-source and free to use.

The aim of this chapter is not to teach how to use MATLAB, but rather to provide a list of useful commands and aspects that should be noted. In order to learn how to use MATLAB, it is recommended to take a course or read a tutorial on the subject. There are many resources available online, including official documentation and user forums, that can help you get started with MATLAB. Additionally, there are many books and online courses that can provide a more in-depth understanding of the language and its capabilities.

- MATLAB starts counting from 1.
- To access the last element of a vector, you can use the command `end`.
- When using vectors, the component-wise operations as `.*` for multiplication, `./` for division, and `.^` for exponentiation are used.
- Different files can be used. Variables can be kept in `.mat` files, while functions can be defined in `.m` files.
- When using a `switch` statement, if it enters a case, it will not execute the following cases, even if they are true. This is different from other programming languages where the execution continues until a `break` statement is encountered.
- Vectorization is a powerful technique in MATLAB that allows you to perform operations on entire arrays or matrices without the need for explicit loops.
- In order to count the time taken by a piece of code, you can use the `tic` and `toc` commands. The `tic` command starts a timer, and the `toc` command stops the timer and returns the elapsed time.
- Comparing vectors can have several effects depending on the context.
 - When comparing two vectors of the same size and shape, the comparison will be performed element-wise, resulting in a new vector of the same size

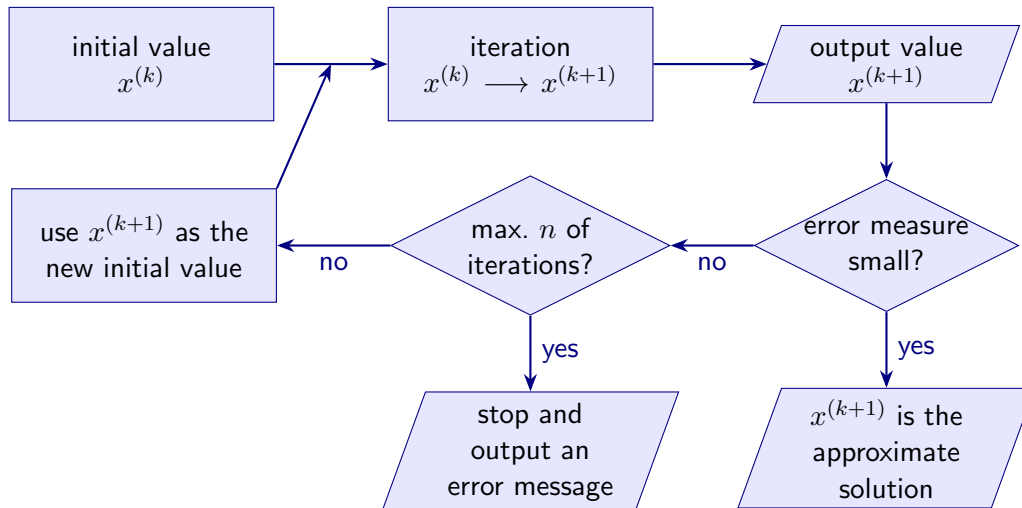


Figura 1.1: General structure of an iterative method.

and shape, where each element is the result of the comparison between the corresponding elements of the original vectors.

- When comparing a vector to a scalar, the comparison will be performed element-wise, resulting in a new vector of the same size and shape as the original vector, where each element is the result of the comparison between the corresponding element of the original vector and the scalar value.
- When comparing two vectors of the same size but opposite directions, the first vector will be compared to the second vector value by value.

1.1. Iterative Methods in MATLAB

An iterative method is a mathematical procedure that generates a sequence of approximations to the solution of a problem, with the goal of converging to the exact solution. The general structure of an iterative method can be represented as shown in Figure 1.1.

1.1.1. Floating Point Arithmetic

In order to work with numerical methods, it is important to understand how floating point arithmetic works. Floating point numbers are a way to represent real numbers in a computer with a fixed number of bits, using the representation shown in Equation 1.1.

$$x = \pm m \times 10^e \quad (1.1)$$

Where:

- m is the mantissa
- e is the exponent, which is an integer that can be positive or negative.
- The sign of the number is determined by the $\pm$ symbol.

That way, a computer can only represent a finite number of real numbers $\mathbb{M} \subset \mathbb{R}$. This leads to several issues, such as:

- Rounding errors: When $x \in \mathbb{R} \setminus \mathbb{M}$, it is approximated by a number in $\mathbb{M}$, which can lead to rounding errors.
- Catastrophic cancellation: When subtracting two nearly equal numbers, the significant digits can be lost, resulting in a large relative error.
- Overflow and underflow: When the result of a calculation exceeds the maximum representable value, it can lead to overflow, while when the result is smaller than the minimum representable value, it can lead to underflow.

This should be taken into account when calculating the error of an iterative method, as it can affect the convergence and accuracy of the method.

- Absorbing elements: When adding a very small number to a very large number, the small number may not affect the result due to the limitations of floating point representation, leading to an absorbing element.

1.1.2. Iterative Methods for LES

In this section, we will discuss some methods that can be used to solve linear equations systems (LES). The reader may have thought of using Gaussian elimination or the inverse of the coefficient matrix, but these methods are not suitable for large systems. The MATLAB command `A\b` automatically chooses the best method iterative method to solve the system $Ax = b$. Even though iterative methods usually take longer to calculate the solution, they are ideal for:

- Sparse systems: When the coefficient matrix A has a large number of zero elements, iterative methods can take advantage of this sparsity to reduce the computational cost.
- Bad conditioned systems: Usual direct methods are really sensitive to errors in the data, while iterative methods can be more robust in these cases.

Condition Number

Given a matrix A , if A is singular (i.e., it does not have an inverse), its condition number $\text{cond}(A)$ is defined as ∞ . If A is non-singular, its condition number is defined as:

$$\text{cond}(A) = \|A\| \cdot \|A^{-1}\| \geq \|A \cdot A^{-1}\| = \|I\| = 1$$

The condition number of a matrix is a measure of how sensitive the solution of a linear system is to changes in the input data. If $\text{cond}(A) \approx 1$, the system is well-conditioned, meaning that small changes in the input data will result in small changes in the solution. If $\text{cond}(A) \gg 1$, the system is bad-conditioned, meaning that small changes in the input data can result in large changes in the solution. The condition number can be used to calculate how good a solution is.

Proposición 1.1. *Let A be a non-singular matrix, and let x be the exact solution of the system $Ax = b$, and $\tilde{x}$ be an approximate solution. Then, the relative error of the solution can be bounded by:*

$$\frac{\|x - \tilde{x}\|}{\|x\|} \leq \text{cond}(A) \cdot \frac{\|r\|}{\|b\|}$$

Where $r = b - A\tilde{x}$ is the residual of the approximate solution.

Demostración.

$$\begin{aligned} \|x\| \cdot \|r\| \cdot \text{cond}(A) &= \|x\| \cdot \|r\| \cdot \|A\| \cdot \|A^{-1}\| \geq \\ &\geq \|Ax\| \cdot \|A^{-1}r\| = \|b\| \cdot \|A^{-1}(b - A\tilde{x})\| = \|b\| \cdot \|x - \tilde{x}\| \end{aligned}$$

Dividing both sides by $\|x\| \cdot \|b\| \geq 0$ gives the desired result. $\square$

2. Vectors and Matrices

This chapter is not introduced, as the reader should be familiar with it. The definitions, properties and how to calculate the following topics should be known:

- Scalar product, cross product, scalar triple product, matrix product
- Linear dependence or independence and orthogonality
- Solvability of linear systems of equations

3. Introduction to calculating with Physical Quantities

Definición 3.1 (Quantity). A quantity is a property of a phenomenon, body or substance that may be distinguished qualitatively and determined quantitatively. It is expressed as the *product of a numerical value and a unit*.

There are two types of quantities:

Scalar A quantity that is completely described by its numerical value and unit. Examples include mass, time, temperature, etc.

Vectorial A quantity that is completely described by its numerical value (module), unit and direction. Examples include velocity, force, etc.

Usually, only the module of a vectorial quantity is given, and the direction is determined by the context. Other times, the direction is given by a unit vector, which is a vector of module 1 whose only purpose is to indicate the direction of the quantity.

In order to work with quantities, we need to introduce the concept of coordinate system.

3.1. Coordinate Systems

A coordinate system of $\mathbb{R}^n$ is defined by a point O called the origin and a base $\mathcal{B} = \{\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n\}$ of $\mathbb{R}^n$. The base vectors $\vec{e}_i$ are called the base vectors of the coordinate system. The coordinates of a point P in the coordinate system are the coefficients of the linear combination of the base vectors that gives the vector $\vec{OP}$:

$$\vec{OP} = \sum_{i=1}^n x_i \vec{e}_i$$

The basic coordinate system is the cartesian coordinate system, which is defined by the origin $O = (0, 0, 0)$ and the base $\mathcal{B}_u = \{\hat{i}, \hat{j}, \hat{k}\}$, where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are the unit vectors in the direction of the x , y and z axes, respectively. They should always follow the right-hand rule:

$$\hat{i} \times \hat{j} = \hat{k}$$

However, there are other coordinate systems that are more convenient to work with in certain situations. In those cases, they are usually not given by a base, but by

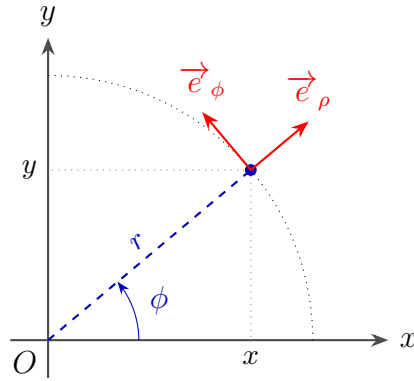


Figura 3.1: Polar coordinates.

a transformation that relates the coordinates of the new system with the coordinates of the cartesian system. In those cases, each vector of the new system is given by fixing all the coordinates of the new system except one, and letting that coordinate vary. More formally, if the new system is given by the coordinates (u_1, u_2, u_3) , then the vector $\vec{e}_i$ of the new system is given by:

$$\vec{e}_i = \frac{\partial \vec{r}}{\partial u_i}$$

where $\vec{r}$ is the position vector of a point.

3.1.1. Polar coordinates

The polar coordinate system (two dimensions) is given by the coordinates (ρ, ϕ) , where ρ is the distance from the origin and ϕ is the angle between the positive x axis and the line that connects the origin with the point. It is shown in Figure 3.1. The transformation that relates the polar coordinates with the cartesian coordinates is given by:

$$\begin{cases} x = \rho \cos \phi \\ y = \rho \sin \phi \end{cases}$$

In order to calculate the base vectors of the polar coordinate system, we need to calculate the position vector of a point in polar coordinates. The position vector of a point P in polar coordinates is given by:

$$\vec{r} = x\hat{i} + y\hat{j} = \rho \cos \phi \hat{i} + \rho \sin \phi \hat{j}$$

Therefore:

$$\begin{aligned} \vec{e}_\rho &= \frac{\cos \phi \hat{i} + \sin \phi \hat{j}}{\sqrt{\cos^2 \phi + \sin^2 \phi}} = \cos \phi \hat{i} + \sin \phi \hat{j} \\ \vec{e}_\phi &= \frac{-\rho \sin \phi \hat{i} + \rho \cos \phi \hat{j}}{\sqrt{(-\rho \sin \phi)^2 + (\rho \cos \phi)^2}} = -\sin \phi \hat{i} + \cos \phi \hat{j} \end{aligned}$$

As we can see, the base vectors of the polar coordinate system are not constant, but they depend on the angle ϕ . This is because the polar coordinate system is

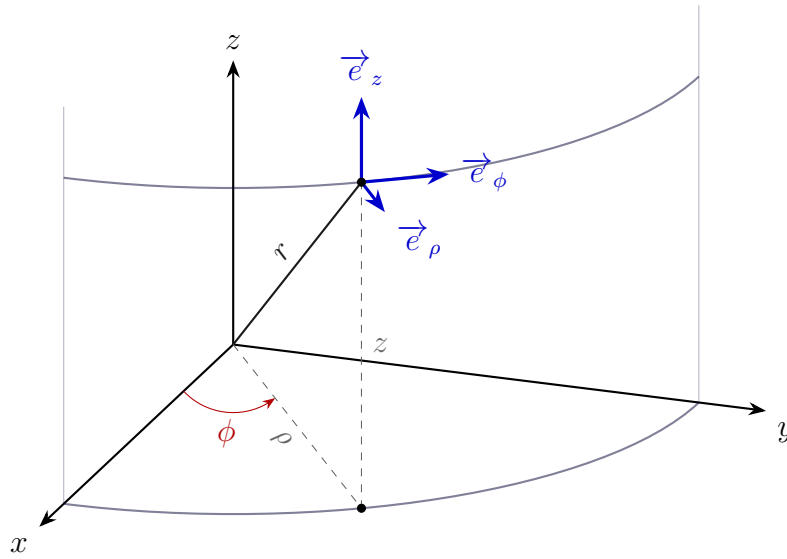


Figura 3.2: Cylindrical coordinates.

a curvilinear coordinate system, where the axes are not straight lines, but curves. Therefore, the base vectors of a curvilinear coordinate system are not constant, but they depend on the coordinates of the point.

3.1.2. Cylindrical coordinates

The cylindrical coordinate system (three dimensions) is given by the coordinates (ρ, ϕ, z) , where ρ and ϕ are the same as in the polar coordinate system when projected onto the xy plane, and z is the height above the xy plane. They are shown in Figure 3.2. The transformation that relates the cylindrical coordinates with the cartesian coordinates is given by:

$$\begin{cases} x = \rho \cos \phi \\ y = \rho \sin \phi \\ z = z \end{cases}$$

The base vectors of the cylindrical coordinate system are given by:

$$\begin{aligned} \vec{e}_\rho &= \cos \phi \hat{i} + \sin \phi \hat{j} \\ \vec{e}_\phi &= -\sin \phi \hat{i} + \cos \phi \hat{j} \\ \vec{e}_z &= \hat{k} \end{aligned}$$

3.1.3. Spherical coordinates

The spherical coordinate system (three dimensions) is given by the coordinates (r, θ, ϕ) , where r is the distance from the origin, θ is the angle between the positive z axis and the line that connects the origin with the point, and ϕ is the angle between the positive x axis and the projection of the line that connects the origin with the

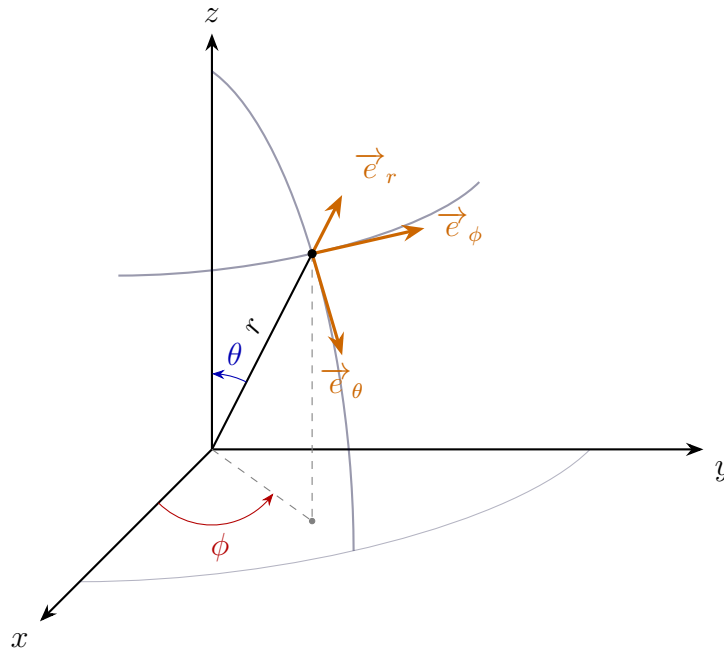


Figura 3.3: Spherical coordinates.

point onto the xy plane. It is shown in Figure 3.3. The transformation that relates the spherical coordinates with the cartesian coordinates is given by:

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases}$$

The base vectors of the spherical coordinate system are given by:

$$\begin{aligned} \vec{e}_r &= \sin \theta \cos \phi \hat{i} + \sin \theta \sin \phi \hat{j} + \cos \theta \hat{k} \\ \vec{e}_\theta &= \cos \theta \cos \phi \hat{i} + \cos \theta \sin \phi \hat{j} - \sin \theta \hat{k} \\ \vec{e}_\phi &= -\sin \phi \hat{i} + \cos \phi \hat{j} \end{aligned}$$

4. Complex Numbers

Complex numbers are an extension of the real numbers, allowing us to work with quantities that have both a real and an imaginary part. A complex number is typically expressed in the form $z = a + bi$, where:

- $a = \Re(z)$ is the real part of the complex number.
- $b = \Im(z)$ is the imaginary part of the complex number.
- i is the imaginary unit, defined by the property $i^2 = -1$.

They can be expressed using different notations:

Cartesian/Algebraic Form: $z = a + bi$

Polar/Trigonometric Form: $z = r(\cos \theta + i \sin \theta)$

The idea behind the polar form is to use polar coordinates to represent the point (a, b) in the complex plane. Here, r is the magnitude (or modulus) of the complex number, and θ is the argument (or angle) of the complex number, measured from the positive real axis.

Exponential Form: $z = re^{i\theta}$

Using the Euler's formula (4.1) we can express the complex number in exponential form, which is often more convenient for multiplication and division of complex numbers.

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (4.1)$$

Using the exponential form, the Moivre's theorem (4.2) can be easily derived, which states that for any complex number $z = re^{i\theta}$ and any integer n :

$$z^n = r^n (\cos(n\theta) + i \sin(n\theta)) = r^n e^{in\theta} \quad (4.2)$$

The reader should be familiar with the basic operations on complex numbers and with the different representations of complex numbers, as well as with the properties of the complex plane.

5. Applications of Complex Numbers

Complex numbers have a wide range of applications in various fields of science and engineering. In the following sections, we will explore some of the most common ones.

5.1. Harmonic Oscillations

6. Übungs Blätter

6.1. Einführung in MATLAB

These exercises are about introducing MATLAB as a tool for numerical computations. They can be found [here](#).

6.2. Grundlagen der Vektor- und Matrizenrechnung

Ejercicio 6.2.1 (Vektoroperationen). Gegeben die Vektoren $\vec{a} = \begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} -1 \\ 5 \\ 8 \end{pmatrix}$ und $\vec{c} = \begin{pmatrix} 5 \\ 0 \\ 1 \end{pmatrix}$. Berechnen Sie die Komponenten (Vektorkoordinaten) sowie Betrag und Richtungswinkel des Vektors

$$\vec{s} = -3\vec{b} + 4(\vec{a} \cdot \vec{c})\vec{a} - (\vec{a} - \vec{b}) \cdot (2\vec{c} + \vec{a})\vec{c}$$

Ejercicio 6.2.2 (Produkte mit Vektoren). Gegeben die Vektoren $\vec{a} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}$ und $\vec{c} = \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix}$. Berechnen Sie mit diesen Vektoren die folgenden Produkte:

1. $\vec{a} \cdot \vec{b}$
2. $(\vec{a} - \vec{c}) \cdot (\vec{a} + \vec{c})$
3. $\vec{b} \times \vec{c}$
4. $(\vec{a} \times \vec{c}) \cdot \vec{b}$

Ejercicio 6.2.3 (Lineare (Un-)Abhängigkeit). Stellen Sie fest, ob die Vektoren $\vec{a}$, $\vec{b}$ und $\vec{c}$ linear unabhängig oder linear abhängig sind:

1. $\vec{a} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}$ und $\vec{c} = \begin{pmatrix} -2 \\ 4 \\ 1 \end{pmatrix}$
2. $\vec{a} = \begin{pmatrix} 5 \\ 1 \\ 1 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix}$ und $\vec{c} = \begin{pmatrix} -13 \\ -2 \\ 13 \end{pmatrix}$

Ejercicio 6.2.4 (Orthogonalität von Vektoren). Eine Kraft $\vec{F} = \begin{pmatrix} F_x \\ F_y \\ F_z \end{pmatrix}$ vom Betrag $F = 100\sqrt{14}$ N soll senkrecht auf einer Ebene E stehen, die durch die Vektoren $\vec{a} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ und $\vec{b} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ aufgespannt wird. Bestimmen Sie die (skalaren) Komponenten F_x , F_y und F_z dieser Kraft. Wieviele Lösungen gibt es?

Ejercicio 6.2.5 (Matrizenmultiplikation).

1. Zeigen Sie am Beispiel der folgenden beiden Matrizen

$$A = \begin{pmatrix} -6 & 9 & -3 \\ -4 & 6 & -2 \\ -2 & 3 & -1 \end{pmatrix} \quad \text{und} \quad B = \begin{pmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{pmatrix},$$

dass die Matrizenmultiplikation nicht-kommutativ ist, d.h. $A \cdot B \neq B \cdot A$.

2. Gegeben die folgenden 3 Matrizen:

$$A_{(3,2)} = \begin{pmatrix} 1 & 3 \\ 2 & 0 \\ 1 & 4 \end{pmatrix}, \quad B_{(2,2)} = \begin{pmatrix} 2 & 1 \\ 3 & -2 \end{pmatrix} \quad \text{und} \quad C_{(2,3)} = \begin{pmatrix} 2 & 0 & -1 \\ 6 & 1 & 3 \end{pmatrix}.$$

- a) Bilden Sie alle möglichen Produkte $X \cdot Y$ aus zwei Faktoren.
b) Welche Produkte $X \cdot Y \cdot Z$ aus drei verschiedenen Faktoren können Sie damit bilden?

Ejercicio 6.2.6 (Lösungsmengen linearer Gleichungssysteme). Bestimmen Sie die Lösungsmenge des Gleichungssystems $A \vec{x} = \vec{b}$ mit

$$A = \begin{pmatrix} 1 & -1 & 2 & 1 & 0 \\ -2 & 4 & -2 & -1 & 3 \\ 0 & -2 & -2 & -1 & -3 \\ 1 & -2 & 2 & 1 & 1 \end{pmatrix}, \quad \vec{b} = \begin{pmatrix} 1 \\ 4 \\ -6 \\ 1 \end{pmatrix}.$$

Was können Sie zur Lösbarkeit des Gleichungssystems sagen?

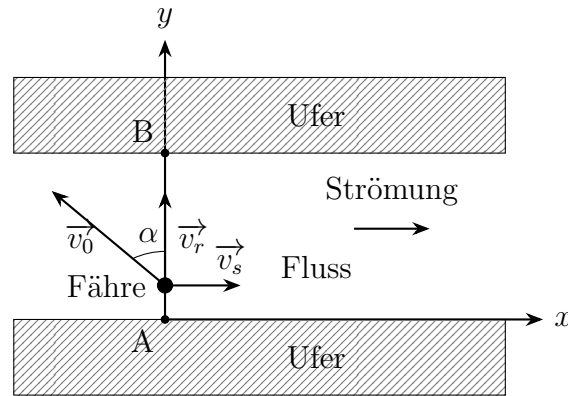


Figura 6.1: Fährrverbindung über einen Fluss mit Strömung.

6.3. Rechnen mit physikalischen Größen

Ejercicio 6.3.1 (Fährverbindung). Eine Fähre bewegt sich mit der Eigengeschwindigkeit $v_0 = 4 \text{ m/s}$ (relativ zum Fluss) vom Uferpunkt A aus auf kürzestem Weg zum gegenüberliegenden Flussufer (Punkt B; Abb. 6.1).

1. Unter welchem Winkel α muss die Fähre gegen die Strömung gesteuert werden, wenn die Geschwindigkeit der Strömung den Betrag $v_s = 1,5 \text{ m/s}$ hat?

The Ferry wants to get to B in an straight line. Therefore, it must compensate the flow velocity by steering at an angle α against the flow. It is then needed that:

$$v_s = v_0 \sin(\alpha) \implies \alpha = \arcsin\left(\frac{v_s}{v_0}\right) = \arcsin\left(\frac{1,5}{4}\right) \approx 0,3844 \text{ rad} \approx 22,024^\circ$$

2. Wie groß ist dann die resultierende Geschwindigkeit v_r der Fähre?

The resulting velocity v_r of the ferry is given by:

$$v_r = v_0 \cos(\alpha) = 4 \cos(0,3844) \approx 3,7081 \text{ m/s}$$

Ejercicio 6.3.2 (Elektrische Punktladungen im Koordinatensystem). Eine Punktladung $q_1 = 6,0 \mu\text{C}$ befindet sich in einem kartesischen Koordinatensystem bei $x_1 = 1,0 \text{ m}$, $y_1 = 0,5 \text{ m}$. Eine zweite Ladung $q_2 = -2,5 \mu\text{C}$ befindet sich in dessen Ursprung. Ein Elektron, d.h. eine dritte Punktladung, ist in einem Punkt mit den Koordinaten (x_e, y_e) . Berechnen Sie die Werte für x_e und y_e , bei denen sich das Elektron im Gleichgewicht befindet, d.h. bei dem die Gesamtkraft auf das Elektron verschwindet.

Option 1

The forces acting on the electron due to the other two charges must cancel each other out for equilibrium. Let F_{ei} be the force on the electron due to charge q_i . The forces can be expressed as:

$$\vec{F}_{e1} + \vec{F}_{e2} = 0$$

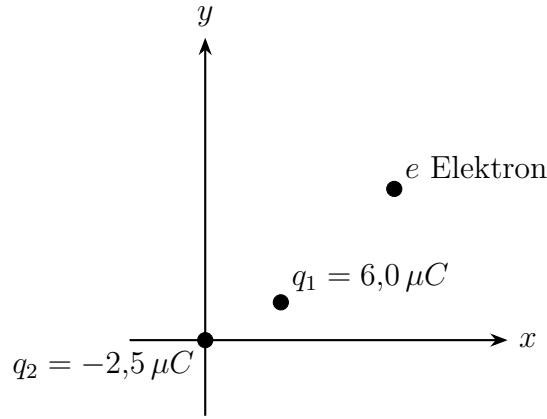


Figura 6.2: Punktladungen im Koordinatensystem.

where, using that $q_e = -e < 0$ (the charge of the electron):

$$\vec{F}_{e1} = -k_e \frac{|q_1 e|}{r_{e1}^2} \hat{r}_{e1}, \quad \vec{F}_{e2} = k_e \frac{|q_2 e|}{r_{e2}^2} \hat{r}_{e2}$$

Here, k_e is Coulomb's constant, e is the elementary charge, r_{ei} is the distance between the electron and charge q_i , and $\hat{r}_{ei}$ is the unit vector pointing from charge q_i to the electron. Using the values of r_{e1} and r_{e2} based on the coordinates of the charges and the electron, we have:

$$\vec{r}_{e1} = (x_e - 1, y_e - 0,5), \quad \vec{r}_{e2} = (x_e, y_e)$$

Therefore, the equilibrium condition becomes:

$$-k_e \frac{|q_1 e|}{((x_e - 1)^2 + (y_e - 0,5)^2)} \cdot \frac{(x_e - 1, y_e - 0,5)}{\sqrt{(x_e - 1)^2 + (y_e - 0,5)^2}} + k_e \frac{|q_2 e|}{(x_e^2 + y_e^2)} \cdot \frac{(x_e, y_e)}{\sqrt{x_e^2 + y_e^2}} = 0$$

This vector equation can be separated into its x and y components, leading to a system of two equations with two unknowns (x_e and y_e). Solving this system will yield the coordinates of the electron in equilibrium.

$$\begin{aligned} -k_e \frac{|q_1 e|(x_e - 1)}{((x_e - 1)^2 + (y_e - 0,5)^2)^{3/2}} + k_e \frac{|q_2 e|x_e}{(x_e^2 + y_e^2)^{3/2}} &= 0 \\ -k_e \frac{|q_1 e|(y_e - 0,5)}{((x_e - 1)^2 + (y_e - 0,5)^2)^{3/2}} + k_e \frac{|q_2 e|y_e}{(x_e^2 + y_e^2)^{3/2}} &= 0 \end{aligned}$$

In an easier way:

$$\begin{aligned} \frac{|q_1|(x_e - 1)}{((x_e - 1)^2 + (y_e - 0,5)^2)^{3/2}} &= \frac{|q_2|x_e}{(x_e^2 + y_e^2)^{3/2}} \\ \frac{|q_1|(y_e - 0,5)}{((x_e - 1)^2 + (y_e - 0,5)^2)^{3/2}} &= \frac{|q_2|y_e}{(x_e^2 + y_e^2)^{3/2}} \end{aligned}$$

As we can see, this system of equations is nonlinear and may require numerical methods or iterative approaches to solve for x_e and y_e . We should then approach the problem using another method.

Option 2

Given that $\vec{F}_{e1} = -\vec{F}_{e2}$, the forces must have the same magnitude but opposite directions. This implies that the electron must lie along the line connecting the two charges, with $x_e, y_e < 0$. Therefore:

$$y_e = \frac{y_1}{x_1} x_e = \frac{1}{2} x_e$$

On the other hand, equalizing the magnitudes of the forces:

$$|\vec{F}_{e1}| = |\vec{F}_{e2}| \implies k_e \frac{|q_1 e|}{r_{e1}^2} = k_e \frac{|q_2 e|}{r_{e2}^2} \implies \frac{|q_1|}{r_{e1}^2} = \frac{|q_2|}{r_{e2}^2} \implies \frac{|q_1|}{(x_e - 1)^2 + (y_e - 0,5)^2} = \frac{|q_2|}{x_e^2 + y_e^2}$$

Therefore, using the relation for y_e in terms of x_e , we have the following equation to solve for x_e :

$$\begin{aligned} \frac{|q_1|}{(x_e - 1)^2 + (\frac{1}{2}x_e - 0,5)^2} &= \frac{|q_2|}{x_e^2 + (\frac{1}{2}x_e)^2} \\ 6 \cdot \left(\frac{5}{4}x_e^2\right) &= 2,5 \cdot \left(\frac{5}{4}x_e^2 - 2,5x_e + 1,25\right) \\ 4,375x_e^2 + 6,25x_e - 3,125 &= 0 \\ x_e &= \frac{-6,25 \pm \sqrt{(-6,25)^2 - 4 \cdot 4,375 \cdot (-3,125)}}{2 \cdot 4,375} = \frac{-5 \pm 2\sqrt{15}}{7} \end{aligned}$$

Given that $x_e < 0$, we take:

$$\begin{aligned} x_e &= -\frac{5 + 2\sqrt{15}}{7} \approx -1,8208 \text{ m} \\ y_e &= \frac{1}{2}x_e = -\frac{5 + 2\sqrt{15}}{14} \approx -0,9104 \text{ m} \end{aligned}$$

Ejercicio 6.3.3 (Die magnetische Kraft). Ein punktförmiges Teilchen mit einer Ladung $q = -3,64 \text{ nC}$ bewegt sich mit einer Geschwindigkeit $\vec{v} = 2,75 \cdot 10^6 \text{ m/s } \vec{e}_x$, d.h. entlang der x-Achse. Berechnen Sie die Kraft, die folgende Felder auf das Teilchen ausüben:

1. $\vec{B} = 0,38 \text{ T } \vec{e}_y$

$$\begin{aligned} \vec{F} &= q \cdot (\vec{v} \times \vec{B}) = q \cdot \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ v_x & 0 & 0 \\ 0 & B_y & 0 \end{vmatrix} = q \cdot v_x \cdot B_y \cdot \vec{e}_z = \\ &= -3,64 \cdot 10^{-9} \cdot 2,75 \cdot 10^6 \cdot 0,38 \vec{e}_z = -3,8038 \cdot 10^{-3} \text{ N } \vec{e}_z \end{aligned}$$

2. $\vec{B} = 0,75 \vec{e}_x + 0,75 \vec{e}_y$

$$\begin{aligned} \vec{F} &= q \cdot (\vec{v} \times \vec{B}) = q \cdot \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ v_x & 0 & 0 \\ B_x & B_y & 0 \end{vmatrix} = q \cdot v_x \cdot B_y \cdot \vec{e}_z = \\ &= -3,64 \cdot 10^{-9} \cdot 2,75 \cdot 10^6 \cdot 0,75 \vec{e}_z = -7,5075 \cdot 10^{-3} \text{ N } \vec{e}_z \end{aligned}$$

6.4. Komplexe Zahlen

Ejercicio 6.4.1 (Eine Rechenaufgabe mit komplexen Zahlen). Gegeben seien drei komplexe Zahlen $z_1 = 2 - i$, $z_2 = 5 + 2i$ und $z_3 = -3i$. Berechne:

$$\frac{\overline{z_1} + 2z_3 \cdot z_1 - iz_2}{2(z_2 - \overline{z_3}) \cdot z_1}$$

Ejercicio 6.4.2 (Rechnen mit komplexen Zahlen).

1. Berechnen Sie Realteil, Imaginärteil und Betrag der komplexen Zahl

$$z = 3 \cdot \overline{1 + 2i} \cdot (5 + i) + 5 \cdot \frac{28i - 3}{14 + (1 + 2i)^3}.$$

2. Bestimmen Sie alle $z \in \mathbb{C}$, die folgende Gleichung lösen:

$$z \cdot \overline{z} + (3 - 4i)z + (3 + 4i)\overline{z} + 9 = 0.$$

Beschreiben Sie die Gestalt der Lösungsmenge in der Gaußschen Zahlenebene.

3. Berechnen Sie Realteil, Imaginärteil, Argument und Betrag von $z \in \mathbb{C}$:

$$z = 2^{-30} \cdot \left(i - \sqrt{3}\right)^{32}.$$

4. Bestimmen Sie Betrag und Argument aller $v \in \mathbb{C}$, die im 2. Quadranten liegen (d.h. $\Re(z) \leq 0$, $\Im(z) \geq 0$) und folgende Gleichung lösen:

$$v^6 = -1 + i.$$

5. Berechnen Sie

$$z = z_1 + z_2 + z_3 = 2 \cdot e^{-5/4i\pi} + 0,2(1 - 3i)^3 + \frac{2 - i}{3 + 4i}.$$

Geben Sie das Ergebnis sowohl in kartesischer als auch in exponentieller Form an.

Ejercicio 6.4.3 (Beträge komplexer Zahlen). Beweisen oder widerlegen Sie: Für alle komplexen Zahlen gilt:

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2 \cdot (|z_1|^2 + |z_2|^2).$$

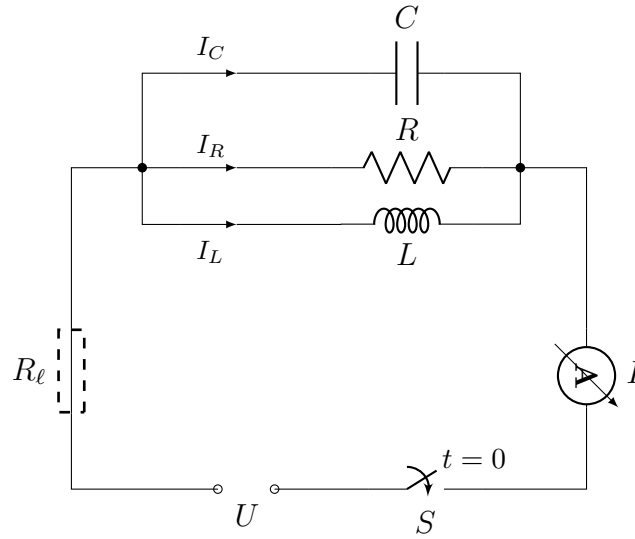


Figura 6.3: Skizze zur Aufgabe 6.5.2

6.5. Anwendung komplexer Rechnung

Ejercicio 6.5.1. Zeigen Sie durch Rechnung in komplexer Form: Drei gleichfrequente, sinusförmige Wechselströme i_1 , i_2 und i_3 mit den Amplituden i_0 , der Kreisfrequenz $\omega > 0$ und den Nullphasenwinkeln $\varphi_1 = 0$, $\varphi_2 = 2/3\pi$ und $\varphi_3 = 4/3\pi$ löschen sich bei ungestörter Überlagerung gegenseitig aus, d.h. $i_1 + i_2 + i_3 = 0$.

Let $i_1(t)$, $i_2(t)$ und $i_3(t)$ be:

$$\begin{aligned} i_1(t) &= i_0 \cdot e^{j(\omega t + 0)} = i_0 \cdot e^{j\omega t} \\ i_2(t) &= i_0 \cdot e^{j(\omega t + 2/3\pi)} = i_0 \cdot e^{j\omega t} \cdot e^{j2/3\pi} \\ i_3(t) &= i_0 \cdot e^{j(\omega t + 4/3\pi)} = i_0 \cdot e^{j\omega t} \cdot e^{j4/3\pi} \end{aligned}$$

Therefore:

$$\begin{aligned} i_1(t) + i_2(t) + i_3(t) &= i_0 \cdot e^{j\omega t} (e^{j0} + e^{j2/3\pi} + e^{j4/3\pi}) = \\ &= i_0 \cdot e^{j\omega t} \left(1 + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} \right) \right) = \\ &= i_0 \cdot e^{j\omega t} \cdot 0 = 0 \end{aligned}$$

Therefore, we have shown that:

$$i_1(t) + i_2(t) + i_3(t) = \Im(i_1(t) + i_2(t) + i_3(t)) = \Im(0) = 0$$

Ejercicio 6.5.2 (Resonanz im Parallelschwingkreis). Der in Abbildung 6.3 skizzierte Parallelschwingkreis mit dem ohmschen Widerstand $R = 10 \Omega$, der Induktivität $L = 0,2 \text{ H}$ und der Kapazität $C = 10 \text{ mF}$ wird durch eine Wechselstromquelle mit dem Effektivwert $I = 10 \text{ A}$ und der variablen Kreisfrequenz ω zu elektromagnetischen Schwingungen angeregt.

1. Im Resonanzfall sind der Gesamtstrom I und die angelegte Spannung U in Phase, d.h. $\varphi_I - \varphi_U = 0$. Bei welcher Kreisfrequenz ω_0 tritt dieser Fall ein? Wie groß ist dann der komplexe Gesamtwiderstand Z ?

The complex resistance is:

$$\underline{Z} = \frac{u(t)}{\underline{i}(t)} = \frac{\sqrt{2}U e^{j(\omega t + \varphi_U)}}{\sqrt{2}I e^{j(\omega t + \varphi_I)}} = \frac{U}{I} e^{j(\varphi_U - \varphi_I)} \stackrel{(*)}{=} \frac{U}{I} e^{j0} = \frac{U}{I} \in \mathbb{R}$$

where in $(*)$ we used the fact that in resonance $\varphi_U = \varphi_I$.

We can calculate the complex resistance as:

$$\begin{aligned} \frac{1}{\underline{Z}} &= \frac{1}{\underline{Z}_R} + \frac{1}{\underline{Z}_L} + \frac{1}{\underline{Z}_C} \\ &= \frac{1}{R} + \frac{1}{j\omega L} + j\omega C \\ \Rightarrow \underline{Z} &= \frac{1}{\frac{1}{R} - j\frac{1}{\omega L} + j\omega C} = \frac{1}{\frac{1}{R} + j\left(-\frac{1}{\omega L} + \omega C\right)} \end{aligned}$$

Given that $\underline{Z} \in \mathbb{R}$, we have that the imaginary part must be zero:

$$\begin{aligned} -\frac{1}{\omega_0 L} + \omega_0 C &= 0 \Rightarrow \omega_0 C = \frac{1}{\omega_0 L} \Rightarrow \omega_0^2 = \frac{1}{LC} \Rightarrow \\ \Rightarrow \omega_0 &= \sqrt{\frac{1}{LC}} = \sqrt{\frac{1}{0,2 \cdot 10 \cdot 10^{-3}}} = \sqrt{500} \approx 22,36 \text{ rad/s} \end{aligned}$$

Therefore, we have that:

$$\underline{Z} = Z = R = 10 \Omega$$

2. Wie ändert sich die Situation, wenn die Zuleitungen nicht mehr als ideal angenommen werden und einen ohmschen Widerstand R_ℓ beitragen? Welche Werte haben ω_0 und Z dann?

In this case, we have:

$$\underline{Z} = \frac{1}{\frac{1}{R} + j\left(-\frac{1}{\omega L} + \omega C\right)} + R_\ell$$

Therefore, ω_0 does not change, but:

$$\underline{Z} = Z = R + R_\ell = 10 + R_\ell$$

Ejercicio 6.5.3 (Wechselstrommessbrücke). Mit der in Abbildung 6.4 dargestellten Brückenschaltung lässt sich ein unbekannter komplexer Widerstand $\underline{Z}_1 = \underline{Z}_X$ wie folgt bestimmen: Bei vorgegebenen (komplexen) Widerständen $\underline{Z}_2$ und $\underline{Z}_3$ wird der stetig veränderbare komplexe Widerstand $\underline{Z}_4$ so eingestellt, dass der Brückenweig A–B stromlos wird. Das in die Brücke geschaltete Wechselstromamperemeter mit dem (bekannten) Innenwiderstand $\underline{Z}_5$ dient dabei lediglich als Nullindikator.

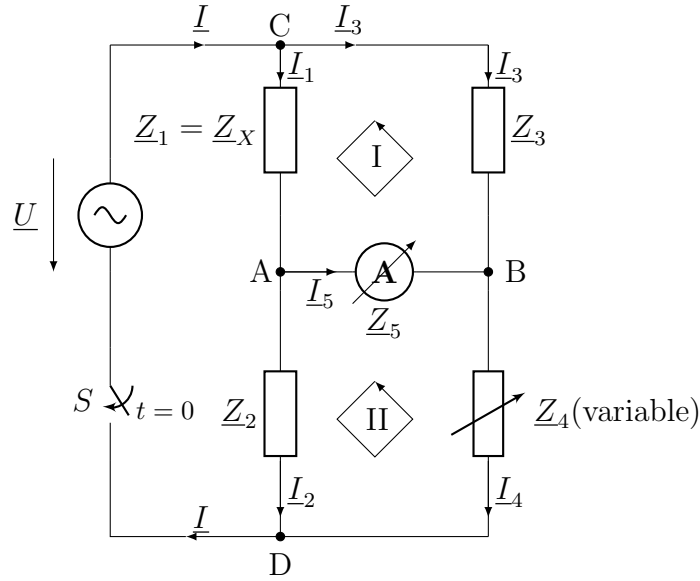


Figura 6.4: Skizze zur Aufgabe 6.5.3.

1. Wie lautet die sogenannte Abgleichbedingung, d.h. die Bedingung für die Stromlosigkeit des Brückenweiges A-B?

Let's denote $U_{ij} := \underline{U}_i - \underline{U}_j$ the voltage between nodes i and j . According to Kirchhoff's laws, we have that:

$$\begin{aligned}\underline{U}_{CA} + \underline{U}_{AB} - \underline{U}_{CB} &= 0 \\ \underline{U}_{AD} - \underline{U}_{BD} - \underline{U}_{AB} &= 0\end{aligned}$$

In order that the branch A-B is currentless, we need:

$$0 = \underline{I}_5 = \frac{\underline{U}_{AB}}{\underline{Z}_5} \iff \underline{U}_{AB} = 0$$

Therefore, we have:

$$\underline{U}_{CA} = \underline{U}_{CB} \quad \underline{U}_{AD} = \underline{U}_{BD} \quad (6.1)$$

Using now Kirchhoff's Intensity Law on nodes A and B , taking into account that $\underline{I}_5 = 0$, we have:

$$\begin{aligned}\underline{I}_1 &= \underline{I}_2 \\ \underline{I}_3 &= \underline{I}_4\end{aligned}$$

Using Ohm's Law, we have:

$$\frac{\underline{U}_{CA}}{\underline{Z}_1} = \frac{\underline{U}_{AD}}{\underline{Z}_2} \quad \frac{\underline{U}_{CB}}{\underline{Z}_3} = \frac{\underline{U}_{BD}}{\underline{Z}_4} \quad (6.2)$$

Unifying Equations (6.1) and (6.2), we obtain:

$$\begin{aligned}\frac{\underline{U}_{CA}}{\underline{Z}_1} = \frac{\underline{U}_{AD}}{\underline{Z}_2} &\implies \frac{\underline{U}_{CA}}{\underline{U}_{AD}} = \frac{\underline{Z}_1}{\underline{Z}_2} \\ \frac{\underline{U}_{CA}}{\underline{Z}_3} = \frac{\underline{U}_{AD}}{\underline{Z}_4} &\implies \frac{\underline{U}_{CA}}{\underline{U}_{AD}} = \frac{\underline{Z}_3}{\underline{Z}_4}\end{aligned}$$

We should denote that we can divide by $\underline{U}_{AD}$ because if $\underline{U}_{AD} = 0$, then $\underline{U}_{BD} = \underline{U}_{AD} = 0$ and therefore $\underline{I}_2 = \underline{I}_4 = 0$, which according to the Kirchhoff's Intensity Law implies that $\underline{I} = 0$, which would mean that $\underline{U} = 0$, which is not the case.

Therefore, we have the balance condition:

$$\frac{\underline{Z}_1}{\underline{Z}_2} = \frac{\underline{Z}_3}{\underline{Z}_4} \implies \underline{Z}_1 \underline{Z}_4 = \underline{Z}_2 \underline{Z}_3$$

2. In einem konkreten Fall haben die festen Widerstände $\underline{Z}_2$ und $\underline{Z}_3$ folgende Werte:

$$\begin{aligned}\underline{Z}_2 &= 10\Omega - j \cdot 2\Omega, \\ \underline{Z}_3 &= 8\Omega - j \cdot 6\Omega.\end{aligned}$$

Die Brücke A–B wird dabei genau dann stromlos, wenn der variable Widerstand $\underline{Z}_4$ auf den Wert

$$\underline{Z}_4 = 5\Omega - j \cdot 2\Omega$$

eingestellt wird. Welchen Wert besitzt dann der (zunächst noch unbekannte) Widerstand $\underline{Z}_X$?

Using the balance condition obtained in the previous part, we obtain:

$$\begin{aligned}\underline{Z}_X &= \underline{Z}_1 = \frac{\underline{Z}_2 \underline{Z}_3}{\underline{Z}_4} \\ &= \frac{(10 - 2j)(8 - 6j)}{5 - 2j} = \\ &= \frac{492}{29}\Omega - j \cdot \frac{244}{29}\Omega \approx 16,97\Omega - j \cdot 8,41\Omega\end{aligned}$$

Ejercicio 6.5.4 (Wechselstromparadoxon). Der in Bild 6.5 dargestellte Wechselstromkreis enthält die ohmschen Widerstände R und R_x und einen zu R_x parallel geschalteten Kondensator mit der Kapazität C . Beim Anlegen einer Wechselspannung $\underline{U}$ mit der Kreisfrequenz ω fließt der Gesamtstrom $\underline{I}$, dessen Effektivwert I durch das zugeschaltete Wechselstrommessgerät A gemessen wird. Der Innenwiderstand R_i des Geräts sei im Widerstand R bereits enthalten. Zeigen Sie: Der ohmsche Widerstand R_x lässt sich so wählen, dass die Stromanzeige unabhängig ist von der Stellung des Schalters S (geschlossen oder offen; sog Wechselstromparadoxon).

We have two options:

- Switch open: In this case, everything is connected in serial and $\underline{I}_1 = 0$. Therefore:

$$\underline{Z}_{\text{open}} = R + \frac{1}{j\omega C} = R - \frac{1}{\omega C} \cdot j$$

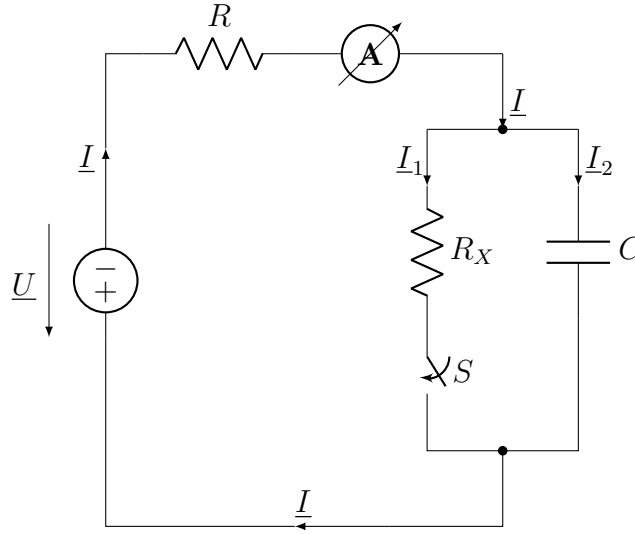


Figura 6.5: Skizze zur Aufgabe 6.5.4.

- Switch closed:

$$\begin{aligned}
 \underline{Z}_{\text{closed}} &= R + \frac{1}{\frac{1}{R_X} + j\omega C} = \left(R + \frac{\frac{1}{R_X}}{\frac{1}{R_X^2} + \omega^2 C^2} \right) - \frac{\omega C}{\frac{1}{R_X^2} + \omega^2 C^2} \cdot j = \\
 &= \left(R + \frac{R_X}{1 + (R_X \omega C)^2} \right) - \frac{R_X^2 \omega C}{1 + (R_X \omega C)^2} \cdot j = \\
 &= \frac{R(1 + (R_X \omega C)^2) + R_X}{1 + (R_X \omega C)^2} - \frac{R_X^2 \omega C}{1 + (R_X \omega C)^2} \cdot j
 \end{aligned}$$

In order that the current does not depend on the position of the switch, it is needed:

$$\hat{I}_{\text{closed}} = \hat{I}_{\text{open}} \implies |\underline{I}_{\text{closed}}| = |\underline{I}_{\text{open}}| \implies \left| \frac{\underline{U}_{\text{closed}}}{\underline{Z}_{\text{closed}}} \right| = \left| \frac{\underline{U}_{\text{open}}}{\underline{Z}_{\text{open}}} \right|$$

Given that $\underline{U}_{\text{open}} = \underline{U}_{\text{closed}}$, we only need that:

$$\begin{aligned}
 |\underline{Z}_{\text{closed}}| &= |\underline{Z}_{\text{open}}| \implies |\underline{Z}_{\text{closed}}|^2 = |\underline{Z}_{\text{open}}|^2 \implies \\
 \implies R^2 + \frac{1}{\omega^2 C^2} &= \frac{(R(1 + (R_X \omega C)^2) + R_X)^2 + (R_X^2 \omega C)^2}{(1 + (R_X \omega C)^2)^2} \implies \\
 \implies (R(1 + (R_X \omega C)^2))^2 + \frac{(1 + (R_X \omega C)^2)^2}{\omega^2 C^2} &= (R(1 + (R_X \omega C)^2))^2 + R_X^2 + 2R_X R(1 + (R_X \omega C)^2) + \\
 \implies \frac{(1 + (R_X \omega C)^2)^2}{\omega^2 C^2} &= R_X^2 + 2R_X R(1 + (R_X \omega C)^2) + (R_X^2 \omega C)^2 \implies \\
 \implies 1 + 2(R_X \omega C)^2 &= (R_X \omega C)^2 + 2\omega^2 C^2 R_X R(1 + (R_X \omega C)^2) \implies \\
 \implies 1 + (R_X \omega C)^2 &= 2\omega^2 C^2 R_X R(1 + (R_X \omega C)^2) \implies \\
 \implies (2\omega^2 C^2 R_X R - 1)(1 + (R_X \omega C)^2) &= 0
 \end{aligned}$$

Therefore, the only possible solution is:

$$R_X = \frac{1}{2R(\omega C)^2}$$

Using that value of R_X , we will measure the same current value with the switch opened and closed.

6.8. Ableitungen von Funktionen mehrerer Veränderlicher

Ejercicio 6.8.1 (Partielle Ableitungen 1. und 2. Ordnung). Bilden Sie alle partiellen Ableitungen erster und zweiter Ordnung der Funktion:

$$\begin{aligned} f : \quad \mathbb{R}^2 &\longrightarrow \mathbb{R} \\ (x, y) &\longmapsto x^2 \cdot e^{-xy} \end{aligned}$$

The partial derivatives of first order are:

$$\begin{aligned} \frac{\partial f}{\partial x}(x, y) &= 2xe^{-xy} - x^2ye^{-xy} = e^{-xy}(2x - x^2y) \\ \frac{\partial f}{\partial y}(x, y) &= -x^3e^{-xy} \end{aligned}$$

The partial derivatives of second order are:

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2}(x, y) &= \frac{\partial}{\partial x} (e^{-xy}(2x - x^2y)) = -ye^{-xy}(2x - x^2y) + e^{-xy}(2 - 2xy) \\ \frac{\partial^2 f}{\partial y^2}(x, y) &= \frac{\partial}{\partial y} (-x^3e^{-xy}) = x^4e^{-xy} \\ \frac{\partial^2 f}{\partial y \partial x}(x, y) &= \frac{\partial}{\partial x} (-x^3e^{-xy}) = -3x^2e^{-xy} + x^3ye^{-xy} \end{aligned}$$

Ejercicio 6.8.2 (Eine Differenzialgleichung). Zeigen Sie, dass die Funktion

$$\begin{aligned} f : \quad \mathbb{R}^2 &\longrightarrow \mathbb{R} \\ (x, t) &\longmapsto e^{-\pi^2 a^2 t} \cdot \sin(\pi x) \end{aligned}$$

eine Lösung der folgenden Gleichung ist:

$$a^2 \frac{\partial^2 f}{\partial x^2} = \frac{\partial f}{\partial t} \quad (a \in \mathbb{R} \text{ const.})$$

Let's compute the partial derivatives:

$$\begin{aligned} \frac{\partial f}{\partial t}(x, t) &= -\pi^2 a^2 e^{-\pi^2 a^2 t} \sin(\pi x) \\ \frac{\partial^2 f}{\partial x^2}(x, t) &= -\pi^2 e^{-\pi^2 a^2 t} \sin(\pi x) \end{aligned}$$

Therefore, it is clear that it is a solution of the given equation.

Ejercicio 6.8.3 (Totales Differenzial einer gebrochen rationalen Funktion). Bestimmen Sie das totale Differenzial der Funktion

$$\begin{aligned} f : \quad \mathbb{R}^2 &\longrightarrow \mathbb{R} \\ (x, y) &\longmapsto \frac{x^2 + y^2}{y - x} \end{aligned}$$

To find the total differential of the function, we first compute the partial derivatives:

$$\begin{aligned}\frac{\partial f}{\partial x}(x, y) &= \frac{2x(y-x) + (x^2 + y^2)}{(y-x)^2} = \frac{-x^2 + 2xy + y^2}{(y-x)^2} \\ \frac{\partial f}{\partial y}(x, y) &= \frac{2y(y-x) - (x^2 + y^2)}{(y-x)^2} = \frac{y^2 - 2xy - x^2}{(y-x)^2}\end{aligned}$$

Therefore the total differential is:

$$\begin{aligned}df(x, y) &= \frac{-x^2 + 2xy + y^2}{(y-x)^2} dx + \frac{y^2 - 2xy - x^2}{(y-x)^2} dy = \\ &= \frac{(-x^2 + 2xy + y^2) dx + (y^2 - 2xy - x^2) dy}{(y-x)^2}\end{aligned}$$

6.9. Kurvenintegrale

Ejercicio 6.9.1 (Kurvenintegral erster Art). Berechnen Sie das Kurvenintegral erster Art $\int_{\varphi_1} f dt$ für die skalarwertige Funktion

$$\begin{aligned} f : \quad \mathbb{R}^3 &\longrightarrow \mathbb{R} \\ (x, y, z) &\longmapsto \sqrt{x^2 + y^2 + z^2} \end{aligned}$$

über die parametrisierte Kurve

$$\begin{aligned} \varphi_1 : [0, 2\pi] &\longrightarrow \mathbb{R}^3 \\ t &\longmapsto (t \cos(t), t \sin(t), t) \end{aligned}$$

First of all, we compute the derivative of the parameterization:

$$\varphi_1'(t) = \begin{pmatrix} \cos(t) - t \sin(t) \\ \sin(t) + t \cos(t) \\ 1 \end{pmatrix}$$

We then realize that ϕ is differentiable, and given that the third component is constant and non-zero, we can conclude that the parameterization is regular. Therefore, in order to compute the norm of the derivative, we have:

$$\begin{aligned} \|\varphi_1'(t)\| &= \sqrt{(\cos(t) - t \sin(t))^2 + (\sin(t) + t \cos(t))^2 + 1^2} \\ &= \sqrt{\cos^2(t) - 2t \sin(t) \cos(t) + t^2 \sin^2(t) + \sin^2(t) + 2t \sin(t) \cos(t) + t^2 \cos^2(t) + 1} \\ &= \sqrt{1 + t^2 + 1} = \sqrt{t^2 + 2} \end{aligned}$$

Therefore, we can now compute the curve integral:

$$\begin{aligned} \int_{\varphi_1} f dt &= \int_0^{2\pi} f(\varphi_1(t)) \|\varphi_1'(t)\| dt \\ &= \int_0^{2\pi} \sqrt{t^2 + t^2} \cdot \sqrt{t^2 + 2} dt \\ &= \sqrt{2} \int_0^{2\pi} t \sqrt{t^2 + 2} dt \end{aligned}$$

We use the substitution given by:

$$\begin{aligned} \Phi : [2, 4\pi^2 + 2] &\longrightarrow [0, 2\pi] \\ u &\longmapsto \sqrt{u - 2} \end{aligned}$$

Thus, using $t = \Phi(u)$, we have:

$$\begin{aligned} \int_{\varphi_1} f dt &= \sqrt{2} \int_{\Phi^{-1}(0)}^{\Phi^{-1}(2\pi)} \Phi(u) \sqrt{\Phi(u)^2 + 2} \cdot \Phi'(u) du \\ &= \sqrt{2} \int_2^{4\pi^2+2} \sqrt{u-2} \sqrt{u} \cdot \frac{1}{2\sqrt{u-2}} du = \frac{\sqrt{2}}{2} \int_2^{4\pi^2+2} \sqrt{u} du = \frac{\sqrt{2}}{2} \left[\frac{2}{3} u^{3/2} \right]_2^{4\pi^2+2} = \\ &= \frac{\sqrt{2}}{3} [u^{3/2}]_2^{4\pi^2+2} = \frac{\sqrt{2}}{3} ((4\pi^2 + 2)^{3/2} - 2^{3/2}) \end{aligned}$$

Ejercicio 6.9.2 (Kurvenintegral zweiter Art). Berechnen Sie das Kurvenintegral zweiter Art $\int_{\varphi_2} \vec{f} d\vec{s}$ für das Vektorfeld

$$\begin{aligned} \vec{f} : \quad \mathbb{R}^3 &\longrightarrow \mathbb{R}^3 \\ (x, y, z) &\longmapsto (x - z, y^2, 1) \end{aligned}$$

über die parametrisierte Kurve

$$\begin{aligned} \varphi_2 : [0, 1] &\longrightarrow \mathbb{R}^3 \\ t &\longmapsto (t^2, t, e^t) \end{aligned}$$

We first compute the derivative of the parameterization:

$$\varphi_2'(t) = (2t, 1, e^t)$$

We then realize that ϕ is differentiable, and given that the third component is non-zero, we can conclude that the parameterization is regular. Then, the dot product $\vec{f}(\varphi_2(t)) \cdot \varphi_2'(t)$ is given by:

$$\begin{aligned} \vec{f}(\varphi_2(t)) \cdot \varphi_2'(t) &= (t^2 - e^t, t^2, 1) \cdot (2t, 1, e^t) \\ &= 2t^3 - 2te^t + t^2 + e^t \end{aligned}$$

Therefore, we can now compute the curve integral:

$$\begin{aligned} \int_{\varphi_2} \vec{f} d\vec{s} &= \int_0^1 \vec{f}(\varphi_2(t)) \cdot \varphi_2'(t) dt \\ &= \int_0^1 (2t^3 - 2te^t + t^2 + e^t) dt \end{aligned}$$

Let's firstly compute the integral of te^t using integration by parts:

$$\begin{bmatrix} u(t) = t & u'(t) = 1 \\ v(t) = e^t & v'(t) = e^t \end{bmatrix}$$

$$\int_0^1 te^t dt = [te^t]_0^1 - \int_0^1 e^t dt$$

Now, we can compute the entire integral:

$$\begin{aligned} \int_{\varphi_2} \vec{f} d\vec{s} &= \left[2 \cdot \frac{t^4}{4} - 2(te^t - e^t) + \frac{t^3}{3} + e^t \right]_0^1 = \left[\frac{t^4}{2} - 2te^t + 3e^t + \frac{t^3}{3} \right]_0^1 = \\ &= \frac{1}{2} - 2e + 3e + \frac{1}{3} - 3 = -\frac{2}{3} + e \end{aligned}$$

Ejercicio 6.9.3 (Länge einer Kurve). Berechnen Sie die Länge der Kurve C mit der Parameterdarstellung

$$\begin{aligned} \varphi : [-1, 1] &\longrightarrow \mathbb{R}^2 \\ t &\longmapsto (x(t), y(t)) \end{aligned}$$

$$\begin{aligned} x : [-1, 1] &\longrightarrow \mathbb{R} \\ t &\longmapsto t^3 \end{aligned}$$

$$\begin{aligned} y : [-1, 1] &\longrightarrow \mathbb{R} \\ t &\longmapsto 1 - t^2 \end{aligned}$$

Observación. Man denke an die Voraussetzung zur Längenberechnung mittels Kurvenintegralen, dass die Parameterdarstellung regulär sein muss, d.h. $\varphi'(t) \neq \vec{0}$. Ist dies nicht für alle Punkte der Kurve der Fall, berechnet man die Länge abschnittsweise.

Given that $x(t)$ is injective, we realize that φ is also injective. Let's compute the derivative of the parameterization:

$$\varphi'(t) = (3t^2, -2t)$$

We realize that φ is differentiable, and that $\varphi'(t) = \vec{0}$ only at $t = 0$. Therefore, we will compute the length of the curve in two parts: in $[-1, 0]$ and in $[0, 1]$. Thus, we compute the norm of the derivative:

$$\|\varphi'(t)\| = \sqrt{(3t^2)^2 + (-2t)^2} = \sqrt{9t^4 + 4t^2} = \sqrt{t^2(9t^2 + 4)} = |t|\sqrt{9t^2 + 4}$$

Therefore, we can now compute the length of the curve:

$$\begin{aligned} L &= \int_{-1}^0 \|\varphi'(t)\| dt + \int_0^1 \|\varphi'(t)\| dt \\ &= \int_{-1}^0 -t\sqrt{9t^2 + 4} dt + \int_0^1 t\sqrt{9t^2 + 4} dt \end{aligned}$$

We use the substitution given by:

$$\begin{aligned} \Phi : \mathbb{R} &\longrightarrow \mathbb{R} \\ t &\longmapsto u = 9t^2 + 4 \end{aligned}$$

In both intervals, Φ is bijective. Thus, using $t = \Phi^{-1}(u)$, we have:

$$\begin{aligned} L &= \int_{\Phi(-1)}^{\Phi(0)} -\Phi^{-1}(u)\sqrt{u} \cdot (\Phi^{-1})'(u) du + \int_{\Phi(0)}^{\Phi(1)} \Phi^{-1}(u)\sqrt{u} \cdot (\Phi^{-1})'(u) du \\ &= \int_{13}^4 -\Phi^{-1}(u)\sqrt{u} \cdot \frac{1}{2\Phi^{-1}(u)} \cdot \frac{1}{9} du + \int_4^{13} \Phi^{-1}(u)\sqrt{u} \cdot \frac{1}{2\Phi^{-1}(u)} \cdot \frac{1}{9} du = \\ &= 2 \cdot \frac{1}{18} \int_4^{13} \sqrt{u} du = \frac{1}{9} \left[\frac{2}{3} u^{3/2} \right]_4^{13} = \frac{2}{27} (13^{3/2} - 4^{3/2}) = \frac{26\sqrt{13} - 16}{27} \end{aligned}$$

Ejercicio 6.9.4 (Flächeninhalt einer Ellipse). Zeigen Sie, dass der Flächeninhalt F einer Ellipse E , die durch die Gleichung

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

beschrieben wird, $F = \pi ab$ ist.

Observación. Verwenden Sie die Parameterdarstellung

$$\begin{aligned} \varphi : [0, 2\pi] &\longrightarrow \mathbb{R}^2 \\ t &\longmapsto (a \cos t, b \sin t) \end{aligned}$$

We first compute the derivative of the parameterization:

$$\varphi'(t) = (-a \sin t, b \cos t)$$

We then realize that ϕ is differentiable, and given that $a, b > 0$ and that both $\sin t$ and $\cos t$ cannot be zero at the same time, we can conclude that the parameterization is regular. Given that the curve goes around the ellipse counter-clockwise, we can use the following formula to compute the area:

$$\begin{aligned} F &= \frac{1}{2} \int_{\varphi} (-y, x) \cdot d\vec{x} = \frac{1}{2} \int_0^{2\pi} (-b \sin t, a \cos t) \cdot (-a \sin t, b \cos t) dt \\ &= \frac{1}{2} \int_0^{2\pi} ab \sin^2 t + ab \cos^2 t dt = \frac{ab}{2} \int_0^{2\pi} 1 dt = \frac{ab}{2} \cdot 2\pi = \pi ab \end{aligned}$$

6.10. Potenzialfelder und Mehrfachintegrale

Ejercicio 6.10.1 (Kurvenintegral einer Funktion mit Potenzial). Im $\mathbb{R}^2$ ist eine Kurve gegeben durch die Parameterdarstellung

$$\begin{aligned}\varphi : [0, \pi/2] &\longrightarrow \mathbb{R}^2 \\ t &\longmapsto (\sin t, t)\end{aligned}$$

Berechnen Sie das Kurvenintegral $\int_{\varphi} \vec{f} \cdot d\vec{s}$ für:

$$\begin{aligned}\vec{f} : \mathbb{R}^2 &\longrightarrow \mathbb{R}^2 \\ (x, y) &\longmapsto \begin{pmatrix} 2xy + 2\cos(y) \\ x^2 - 2x\sin(y) + 1 \end{pmatrix}\end{aligned}$$

Observación. Prüfen Sie, ob $\vec{f}$ ein Potenzial hat und nutzen Sie es ggf. aus.

Let's first check that φ is a regular parametrization:

$$\varphi'(t) = (\cos t, 1)$$

Given that $1 \neq 0$ for all $t \in [0, \pi/2]$, we conclude that φ is regular. Let's now try to calculate the integral directly. In order to do so, we need to compute $\vec{f}(\varphi(t)) \cdot \varphi'(t)$:

$$\begin{aligned}\vec{f}(\varphi(t)) &= \vec{f}(\sin t, t) = \begin{pmatrix} 2\sin t \cdot t + 2\cos t \\ \sin^2 t - 2\sin^2 t + 1 \end{pmatrix} = \begin{pmatrix} 2t\sin t + 2\cos t \\ 1 - \sin^2 t \end{pmatrix} = \begin{pmatrix} 2t\sin t + 2\cos t \\ \cos^2 t \end{pmatrix} \\ \varphi'(t) &= (\cos t, 1)\end{aligned}$$

$$\vec{f}(\varphi(t)) \cdot \varphi'(t) = 2t\sin t \cos t + 2\cos^2 t + \cos^2 t = 2t\sin t \cos t + 3\cos^2 t$$

Therefore, the integral is:

$$\int_{\varphi} \vec{f} \cdot d\vec{s} = \int_0^{\pi/2} (2t\sin t \cos t + 3\cos^2 t) dt$$

Given that integrating this expression is quite cumbersome, let's check if $\vec{f}$ has a potential function. Given that $\mathbb{R}^2$ is star-shaped, we check:

$$\begin{aligned}\frac{\partial f_1}{\partial y} &= 2x - 2\sin(y) \\ \frac{\partial f_2}{\partial x} &= 2x - 2\sin(y)\end{aligned}$$

Since the mixed partial derivatives are equal, $\vec{f}$ has a potential function. Let's find it:

Option 1 Having fixed $(x, y) \in \mathbb{R}^2$, let's consider the following path from $(0, 0)$ to (x, y) :

$$\begin{aligned}\Phi : [0, 1] &\longrightarrow \mathbb{R}^2 \\ t &\longmapsto (tx, ty)\end{aligned}$$

Let's also write the potential function as $u : \mathbb{R}^2 \rightarrow \mathbb{R}$. Then:

$$\int_{\Phi} \vec{f} \cdot d\vec{s} = u(\Phi(1)) - u(\Phi(0)) = u(x, y) \implies u(x, y) = \int_0^1 \vec{f}(\Phi(t)) \cdot \Phi'(t) dt + u(0, 0)$$

Given that the potential is defined up to a constant, we can set $u(0, 0) = 0$.

Now, let's compute $\vec{f}(\Phi(t)) \cdot \Phi'(t)$:

$$\vec{f}(\Phi(t)) = \vec{f}(tx, ty) = \begin{pmatrix} 2(tx)(ty) + 2 \cos(ty) \\ (tx)^2 - 2(tx) \sin(ty) + 1 \end{pmatrix} = \begin{pmatrix} 2t^2xy + 2 \cos(ty) \\ t^2x^2 - 2tx \sin(ty) + 1 \end{pmatrix}$$

$$\Phi'(t) = (x, y)$$

$$\begin{aligned} \vec{f}(\Phi(t)) \cdot \Phi'(t) &= 2t^2x^2y + 2x \cos(ty) + t^2x^2y - 2txy \sin(ty) + y \\ &= 3t^2x^2y + 2x \cos(ty) - 2txy \sin(ty) + y \end{aligned}$$

Therefore, the potential function is:

$$\begin{aligned} u(x, y) &= \int_0^1 (3t^2x^2y + 2x \cos(ty) - 2txy \sin(ty) + y) dt = \\ &= \left[x^2yt^3 + \frac{2x}{y} \sin(ty) + yt \right]_0^1 - 2xy \int_0^1 t \sin(ty) dt = \\ &= x^2y + \frac{2x}{y} \sin(y) + y - 2xy \int_0^1 t \sin(ty) dt \end{aligned}$$

To compute the remaining integral, we use integration by parts:

$$\left[\begin{array}{ll} u(t) = t & u'(t) = 1 \\ v(t) = -\frac{1}{y} \cos(ty) & v'(t) = \sin(ty) \end{array} \right]$$

$$\begin{aligned} \int_0^1 t \sin(ty) dt &= \left[-\frac{t}{y} \cos(ty) \right]_0^1 + \frac{1}{y} \int_0^1 \cos(ty) dt = \\ &= \left[-\frac{t}{y} \cos(ty) + \frac{1}{y^2} \sin(ty) \right]_0^1 = -\frac{1}{y} \cos(y) + \frac{1}{y^2} \sin(y) \end{aligned}$$

Therefore, the potential function is:

$$\begin{aligned} u(x, y) &= x^2y + \frac{2x}{y} \sin(y) + y - 2xy \left(-\frac{1}{y} \cos(y) + \frac{1}{y^2} \sin(y) \right) = \\ &= x^2y + \cancel{\frac{2x}{y} \sin(y)} + y + 2x \cos(y) - \cancel{\frac{2x}{y} \sin(y)} = \\ &= x^2y + y + 2x \cos(y) \end{aligned}$$

Option 2 We look for a function $u(x, y)$ such that:

$$\begin{aligned} \frac{\partial u}{\partial x} &= 2xy + 2 \cos(y) \\ \frac{\partial u}{\partial y} &= x^2 - 2x \sin(y) + 1 \end{aligned}$$

Integrating the first equation with respect to x :

$$u(x, y) = \int 2xy + 2 \cos(y) dx = x^2 y + 2x \cos(y) + h(y)$$

where $h(y)$ is an arbitrary function of y . Now, we differentiate $u(x, y)$ with respect to y and equate it to the second equation:

$$\frac{\partial u}{\partial y} = x^2 - 2x \sin(y) + h'(y) = x^2 - 2x \sin(y) + 1 \implies h'(y) = 1 \implies h(y) = y + C$$

Given that the potential is defined up to a constant, we can set $C = 0$. Therefore, the potential function is:

$$u(x, y) = x^2 y + 2x \cos(y) + y$$

Finally, we can compute the integral using the potential function:

$$\int_{\varphi} \vec{f} \cdot d\vec{s} = u(\varphi(\pi/2)) - u(\varphi(0)) = u(1, \pi/2) - u(0, 0) = \frac{\pi}{2} + 2 \cos(\pi/2) + \frac{\pi}{2} - 0 = \pi$$

Ejercicio 6.10.2 (Doppelintegral mit e-Funktion). Berechnen Sie

$$I = \int_0^1 \int_0^{y^2} e^{x/y} dx dy$$

$$\int_0^1 \int_0^{y^2} e^{x/y} dx dy = \int_0^1 [ye^{x/y}]_0^{y^2} dy = \int_0^1 y(e^y - 1) dy = \int_0^1 ye^y dy - \int_0^1 y dy$$

To compute the remaining integral, we use integration by parts:

$$\begin{bmatrix} u(y) = y & u'(y) = 1 \\ v(y) = e^y & v'(y) = e^y \end{bmatrix}$$

$$\int_0^1 ye^y dy = [ye^y]_0^1 - \int_0^1 e^y dy = [ye^y - e^y]_0^1 = (e - e) - (0 - 1) = 1$$

Therefore, the value of the integral is:

$$I = 1 - \left[\frac{y^2}{2} \right]_0^1 = 1 - \frac{1}{2} = \frac{1}{2}$$

Ejercicio 6.10.3 (Flächenberechnung mit Doppelintegral). Ein Flächenstück wird durch die Kurven $x = 0$, $y = 2x$ $y = 1/ax^2 + a$, $a > 0$ berandet. Berechnen Sie den Flächeninhalt A mit Hilfe eines Doppelintegrals.

In Figure 6.6, we can see the area to be calculated. Let's first find the points of intersection between the curves $y = 2x$ and $y = 1/ax^2 + a$:

$$2x = \frac{1}{a}x^2 + a \implies x^2 - 2ax + a^2 = 0 \implies (x - a)^2 = 0 \implies x = a$$

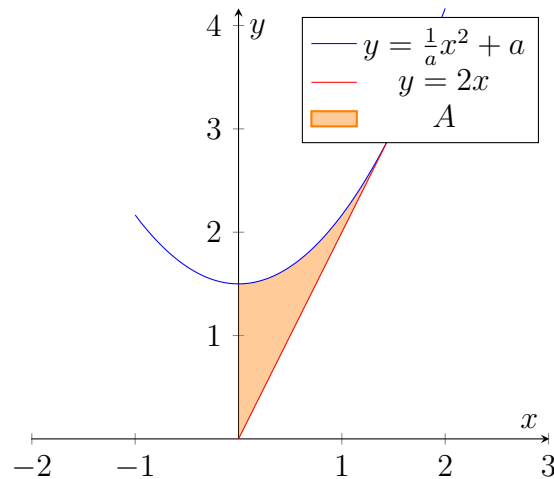


Figura 6.6: Area to be calculated in Exercise 6.10.3.

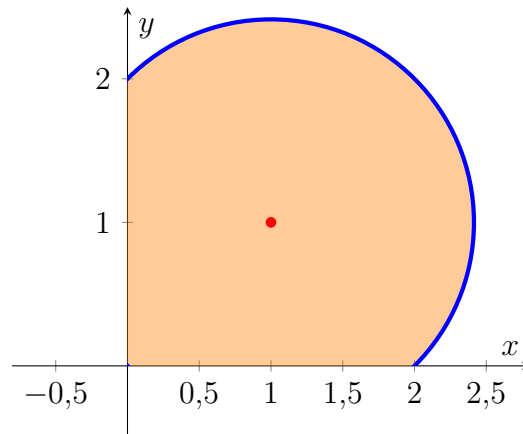


Figura 6.7: Area to be calculated in Exercise 6.10.4.

Therefore, the curves intersect at the point $(a, 2a)$. Given that the area is bounded by $x = 0$ and $x = a$, we can set up the double integral to calculate the area:

$$\begin{aligned} A &= \int_0^a \int_{2x}^{1/ax^2+a} dy \, dx = \int_0^a \left(\frac{1}{a}x^2 + a - 2x \right) dx = \left[\frac{1}{3a}x^3 + ax - x^2 \right]_0^a = \\ &= \frac{1}{3a}a^3 + a^2 - a^2 - 0 = \frac{a^2}{3} \end{aligned}$$

Ejercicio 6.10.4 (Flächeninhalt mit Polarkoordinaten). Die Randkurve eines Gebiets in der Ebene wird durch die Gleichung

$$r = 2(\cos(\varphi) + \sin(\varphi))$$

in Polarkoordinaten beschrieben. Berechnen Sie den Flächeninhalt A dieses Gebiets.

In Figure 6.7, we can see the area to be calculated. Using polar coordinates, the

area can be calculated as:

$$\begin{aligned}
 A &= \int_0^{\pi/2} \int_0^{2(\cos(\varphi)+\sin(\varphi))} r \, dr \, d\varphi = \int_0^{\pi/2} \left[\frac{r^2}{2} \right]_0^{2(\cos(\varphi)+\sin(\varphi))} d\varphi = \int_0^{\pi/2} 2(\cos(\varphi) + \sin(\varphi))^2 d\varphi = \\
 &= 2 \int_0^{\pi/2} (\cos^2(\varphi) + 2\sin(\varphi)\cos(\varphi) + \sin^2(\varphi)) d\varphi = 2 \int_0^{\pi/2} (1 + \sin(2\varphi)) d\varphi = \\
 &= 2 \left[\varphi - \frac{1}{2} \cos(2\varphi) \right]_0^{\pi/2} = 2 \left(\frac{\pi}{2} - \frac{1}{2} \cos(\pi) - \left(0 - \frac{1}{2} \cos(0) \right) \right) = 2 \left(\frac{\pi}{2} + \frac{1}{2} + \frac{1}{2} \right) = \pi + 2
 \end{aligned}$$

Ejercicio 6.10.5 (Dreifachintegral in Zylinderkoordinaten). Berechnen Sie das folgende in Zylinderkoordinaten gegebene Integral:

$$I = \int_{\pi}^{2\pi} \int_0^1 \int_r^{r^2} rz \cdot \sin(\varphi) \, dz \, dr \, d\varphi.$$

$$\begin{aligned}
 I &= \int_{\pi}^{2\pi} \int_0^1 \int_r^{r^2} rz \sin(\varphi) \, dz \, dr \, d\varphi = \int_{\pi}^{2\pi} \int_0^1 \left[\frac{rz^2}{2} \sin(\varphi) \right]_{z=r}^{z=r^2} dr \, d\varphi = \\
 &= \int_{\pi}^{2\pi} \sin(\varphi) \int_0^1 r \cdot \left[\frac{z^2}{2} \right]_r^{r^2} dr \, d\varphi = \int_{\pi}^{2\pi} \sin(\varphi) \int_0^1 r \cdot \left(\frac{r^4}{2} - \frac{r^2}{2} \right) dr \, d\varphi = \\
 &= \frac{1}{2} \int_{\pi}^{2\pi} \sin(\varphi) \int_0^1 (r^5 - r^3) dr \, d\varphi = \frac{1}{2} \int_{\pi}^{2\pi} \sin(\varphi) \left[\frac{r^6}{6} - \frac{r^4}{4} \right]_0^1 d\varphi = \\
 &= \frac{1}{2} \int_{\pi}^{2\pi} \sin(\varphi) \left(\frac{1}{6} - \frac{1}{4} \right) d\varphi = -\frac{1}{24} \int_{\pi}^{2\pi} \sin(\varphi) d\varphi = \frac{1}{24} [\cos(\varphi)]_{\pi}^{2\pi} = \frac{1}{24} (1 - (-1)) = \frac{1}{12}
 \end{aligned}$$

6.11. Datenanalyse und Fehlerrechnung

Ejercicio 6.11.1 (Ein neues Schwimmbecken). Bei einem Schwimmbadbauer wird ein quaderförmiges Becken mit einer Länge $l = 50$ m, einer Breite $b = 15$ m und einer Tiefe $h = 2$ m in Auftrag gegeben. Erfahrungsgemäß baut die Firma ziemlich ungenau mit statistisch verteilten Fehlern von $\sigma_l = 0,4$ m, $\sigma_b = 0,2$ m sowie $\sigma_h = 0,1$ m für Länge, Breite bzw. Tiefe. Berechnen Sie die das Volumen des Schwimmbeckens und den zu erwartenden relativen Fehler nach der Gaußschen Fehlerfortpflanzung. Wieviel Wasser (in m^3) sollte bereitstehen, damit das Becken auf jeden Fall bis zum Rand gefüllt werden kann?

The volume V of the swimming pool is given by the formula:

$$V = l \cdot b \cdot h = 50 \text{ m} \cdot 15 \text{ m} \cdot 2 \text{ m} = 1500 \text{ m}^3$$

To calculate the relative error in the volume using Gaussian error propagation, we use the formula:

$$\begin{aligned} \sigma_V &= \sqrt{\left(\frac{\partial V}{\partial l} \sigma_l\right)^2 + \left(\frac{\partial V}{\partial b} \sigma_b\right)^2 + \left(\frac{\partial V}{\partial h} \sigma_h\right)^2} = \\ &= \sqrt{(b \cdot h \cdot \sigma_l)^2 + (l \cdot h \cdot \sigma_b)^2 + (l \cdot b \cdot \sigma_h)^2} = \\ &= \sqrt{(15 \text{ m} \cdot 2 \text{ m} \cdot 0,4 \text{ m})^2 + (50 \text{ m} \cdot 2 \text{ m} \cdot 0,2 \text{ m})^2 + (50 \text{ m} \cdot 15 \text{ m} \cdot 0,1 \text{ m})^2} = \\ &= \sqrt{(12 \text{ m}^3)^2 + (20 \text{ m}^3)^2 + (75 \text{ m}^3)^2} = \\ &= \sqrt{144 \text{ m}^6 + 400 \text{ m}^6 + 5625 \text{ m}^6} = \\ &= \sqrt{6169 \text{ m}^6} \approx 78,55 \text{ m}^3 \end{aligned}$$

The relative error in the volume is then given by:

$$\frac{\sigma_V}{V} = \frac{78,55 \text{ m}^3}{1500 \text{ m}^3} \approx 0,0524 \approx 5,24 \%$$

Therefore, the volume of the swimming pool is 1500 m^3 with an expected relative error of approximately $5,24 \%$. To ensure that the pool can be filled to the brim with a level of confidence of around $99,7 \%$ (which corresponds to 3σ), we should prepare the following amount of water:

$$V + 3\sigma_V = 1500 \text{ m}^3 + 3 \cdot 78,55 \text{ m}^3 \approx 1735,65 \text{ m}^3$$

Ejercicio 6.11.2 (Selbstinduktion einer Doppelleitung). Eine elektrische Doppelleitung besteht aus zwei parallelen, elektrisch leitenden Drähten mit der Länge l und dem Radius r . Der Abstand ihrer Mittelpunkte beträgt a . Die Selbstinduktivität L dieser Doppelleitung in Luft wird dabei nach der Formel

$$L(l, r, a) = \frac{\mu_0}{\pi} l \cdot \ln \left(\frac{a - r}{r} + \frac{1}{4} \right)$$

berechnet, mit der magnetischen Feldkonstante $\mu_0 = 4\pi \cdot 10^{-7} \text{ Vs/Am}$.

1. Welche Selbstinduktivität L besitzt eine Doppelleitung, deren Dimensionen wie folgt gemessen wurden?

$$l = (2000 \pm 10 \text{ m}); \quad r = (2 \pm 0,05 \text{ mm}); \quad a = (30 \pm 0,3 \text{ cm})$$

The self-inductance L of the twin-lead can be calculated using the given formula:

$$\begin{aligned} L &= \frac{\mu_0}{\pi} l \cdot \ln \left(\frac{a-r}{r} + \frac{1}{4} \right) = \\ &= \frac{4\pi \cdot 10^{-7} \text{ Vs/Am}}{\pi} \cdot 2000 \text{ m} \cdot \ln \left(\frac{0,30 \text{ m} - 0,002 \text{ m}}{0,002 \text{ m}} + \frac{1}{4} \right) = \\ &= 8 \cdot 10^{-4} \text{ Vs/A} \ln(149 + 0,25) \approx 8 \cdot 10^{-4} \text{ Vs/A} \cdot 5,005622 = \\ &= 4,004498 \text{ mH} \end{aligned}$$

2. Wie groß ist die absolute bzw. prozentuale Messunsicherheit des Mittelwertes von L (d. h. der absolute bzw. prozentuale mittlere Fehler des Mittelwertes von L)?

The absolute uncertainty in L can be calculated using Gaussian error propagation:

$$\begin{aligned} \Delta L &= \sqrt{\left(\frac{\partial L}{\partial l} \Delta l \right)^2 + \left(\frac{\partial L}{\partial r} \Delta r \right)^2 + \left(\frac{\partial L}{\partial a} \Delta a \right)^2} = \\ &= \sqrt{\left(\frac{\mu_0}{\pi} \ln \left(\frac{a-r}{r} + \frac{1}{4} \right) \Delta l \right)^2 + \left(\frac{\mu_0}{\pi} l \cdot \frac{1}{\frac{a}{r} - \frac{3}{4}} \cdot \frac{a}{r^2} \Delta r \right)^2 + \left(\frac{\mu_0}{\pi} l \cdot \frac{1}{\frac{a}{r} - \frac{3}{4}} \cdot \frac{1}{r} \Delta a \right)^2} = \\ &= \sqrt{\left(\frac{\mu_0}{\pi} \ln \left(\frac{a}{r} - \frac{3}{4} \right) \Delta l \right)^2 + \left(\frac{\mu_0}{\pi} l \cdot \frac{a}{ar - \frac{3r^2}{4}} \Delta r \right)^2 + \left(\frac{\mu_0}{\pi} l \cdot \frac{1}{a - \frac{3r}{4}} \Delta a \right)^2} \approx \\ &\approx 2,95 \cdot 10^{-5} \text{ H} = \\ &= 0,0295 \text{ mH} \end{aligned}$$

The percentage uncertainty in L is then given by:

$$\frac{\Delta L}{L} = \frac{0,0295 \text{ mH}}{4,004498 \text{ mH}} \approx 0,00737 \approx 0,737 \%$$

Ejercicio 6.11.3 (Regressionsanalyse). Gegeben die folgende Wertetabelle:

t	-2	-1	0	1	2
y	$13,1$	-2	$-2,2$	-4	$-15,9$

Finden Sie x_1 und x_2 für die beste Kurve der Form $y = f(t) = x_1 + x_2 t^3$ zur Anpassung der Messdaten. Welchen Wert hat das zugehörige Bestimmtheitsmaß R^2 ?